

Coding Challenge 7

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```
install.packages("lme4") install.packages("emmeans") install.packages("multcomp") install.packages("multcompView")
```

Question 1

Read in the data called "PlantEmergence.csv" using a relative file path and load the following libraries. tidyverse, lme4, emmeans, multcomp, and multcompView. Turn the Treatment , DaysAfterPlanting and Rep into factors using the function as.factor

```
library(lme4)
```

```
## Loading required package: Matrix
```

```
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.3
```

```
## Welcome to emmeans.
```

```
## Caution: You lose important information if you filter this package's results.
```

```
## See '? untidy'
```

```
library(multcomp)
```

```
## Loading required package: mvtnorm
```

```
## Warning: package 'mvtnorm' was built under R version 4.5.3
```

```
## Loading required package: survival
```

```
## Loading required package: TH.data

## Loading required package: MASS

##
## Attaching package: 'TH.data'

## The following object is masked from 'package:MASS':
##
##      geyser

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.2      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.1

## -- Conflicts ----- tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x tidyr::pack()    masks Matrix::pack()
## x dplyr::select() masks MASS::select()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(multcompView)

## Warning: package 'multcompView' was built under R version 4.5.3

PlantEmergence <- read.csv("PlantEmergence.csv", na.string = "na")
PlantEmergence$Treatment <- as.factor(PlantEmergence$Treatment)
PlantEmergence$Rep <- as.factor(PlantEmergence$Rep)
PlantEmergence$DaysAfterPlanting <- as.factor(PlantEmergence$DaysAfterPlanting)
```

Question 2

Fit a linear model to predict Emergence using Treatment and DaysAfterPlanting along with the interaction. Provide the summary of the linear model and ANOVA results.

Looking at the summary of the linear model, we can see the intercept and plants using Treatment 2 ($p < .001$) have significant effects on emergence. There is, however, no significant difference across any of the other variables. Looking at the ANOVA, we see that days after planting ($p < .001$) and treatment ($p < .001$) have significant effects on plant emergence. There is no significant difference in the interaction of both variables.

```
lm_model <- lm(Emergence ~ DaysAfterPlanting * Treatment, data = PlantEmergence)
summary(lm_model)
```

```
##
## Call:
## lm(formula = Emergence ~ DaysAfterPlanting * Treatment, data = PlantEmergence)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.250  -6.062  -0.875   6.750  21.875
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.823e+02  5.324e+00  34.229  <2e-16 ***
## DaysAfterPlanting14      1.000e+01  7.530e+00   1.328   0.187
## DaysAfterPlanting21      1.062e+01  7.530e+00   1.411   0.161
## DaysAfterPlanting28      1.100e+01  7.530e+00   1.461   0.147
## Treatment2          -1.365e+02  7.530e+00 -18.128  <2e-16 ***
## Treatment3           1.112e+01  7.530e+00   1.477   0.142
## Treatment4           2.500e+00  7.530e+00   0.332   0.741
## Treatment5           8.750e+00  7.530e+00   1.162   0.248
## Treatment6           7.000e+00  7.530e+00   0.930   0.355
## Treatment7          -1.250e-01  7.530e+00  -0.017   0.987
## Treatment8           9.125e+00  7.530e+00   1.212   0.228
## Treatment9           2.375e+00  7.530e+00   0.315   0.753
## DaysAfterPlanting14:Treatment2  1.625e+00  1.065e+01   0.153   0.879
## DaysAfterPlanting21:Treatment2  3.500e+00  1.065e+01   0.329   0.743
## DaysAfterPlanting28:Treatment2  2.750e+00  1.065e+01   0.258   0.797
## DaysAfterPlanting14:Treatment3 -2.625e+00  1.065e+01  -0.247   0.806
## DaysAfterPlanting21:Treatment3 -1.000e+00  1.065e+01  -0.094   0.925
## DaysAfterPlanting28:Treatment3 -1.875e+00  1.065e+01  -0.176   0.861
## DaysAfterPlanting14:Treatment4 -6.250e-01  1.065e+01  -0.059   0.953
## DaysAfterPlanting21:Treatment4  1.500e+00  1.065e+01   0.141   0.888
## DaysAfterPlanting28:Treatment4  3.134e-13  1.065e+01   0.000   1.000
## DaysAfterPlanting14:Treatment5  2.500e+00  1.065e+01   0.235   0.815
## DaysAfterPlanting21:Treatment5  2.875e+00  1.065e+01   0.270   0.788
## DaysAfterPlanting28:Treatment5  2.500e+00  1.065e+01   0.235   0.815
## DaysAfterPlanting14:Treatment6  1.000e+00  1.065e+01   0.094   0.925
## DaysAfterPlanting21:Treatment6  4.125e+00  1.065e+01   0.387   0.699
## DaysAfterPlanting28:Treatment6  2.125e+00  1.065e+01   0.200   0.842
## DaysAfterPlanting14:Treatment7 -2.500e+00  1.065e+01  -0.235   0.815
## DaysAfterPlanting21:Treatment7 -2.125e+00  1.065e+01  -0.200   0.842
## DaysAfterPlanting28:Treatment7 -3.625e+00  1.065e+01  -0.340   0.734
## DaysAfterPlanting14:Treatment8 -2.500e+00  1.065e+01  -0.235   0.815
## DaysAfterPlanting21:Treatment8 -1.500e+00  1.065e+01  -0.141   0.888
## DaysAfterPlanting28:Treatment8 -1.500e+00  1.065e+01  -0.141   0.888
## DaysAfterPlanting14:Treatment9  6.250e-01  1.065e+01   0.059   0.953
## DaysAfterPlanting21:Treatment9 -1.250e+00  1.065e+01  -0.117   0.907
## DaysAfterPlanting28:Treatment9 -8.750e-01  1.065e+01  -0.082   0.935
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.65 on 108 degrees of freedom
## Multiple R-squared:  0.9585, Adjusted R-squared:  0.945
## F-statistic: 71.21 on 35 and 108 DF, p-value: < 2.2e-16
```

```
anova(lm_model)
```

```
## Analysis of Variance Table
##
## Response: Emergence
##              Df Sum Sq Mean Sq  F value    Pr(>F)
## DaysAfterPlanting      3    3116      1039    9.1603 1.877e-05 ***
## Treatment              8 279366     34921 307.9516 < 2.2e-16 ***
## DaysAfterPlanting:Treatment 24     142        6    0.0522      1
## Residuals            108  12247      113
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 3

Based on the results of the linear model in question 2, do you need to fit the interaction term? Provide a simplified linear model without the interaction term but still testing both main effects. Provide the summary and ANOVA results. Then, interpret the intercept and the coefficient for Treatment 2.

Based on the results of question 2, we do need to fit the interaction term to better understand the variables outside of the interaction.

Looking at the summary of the linear model, we can see that there is a significant difference in emergence 14, 21, and 28 days after planting. Additionally, we can see that all treatments had a significant effect on emergence except for treatments 4, 7, and 9. In reviewing the ANOVA table, we see that emergence is significantly affected by days since planting (daysafterplanting; $p < .001$) and treatment received ($p < .001$).

```
lm_model_q3 <- lm(Emergence~DaysAfterPlanting+Treatment, data = PlantEmergence)
summary(lm_model_q3)
```

```
##
## Call:
## lm(formula = Emergence ~ DaysAfterPlanting + Treatment, data = PlantEmergence)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.1632  -6.1536  -0.8542   6.1823  21.3958
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      182.163      2.797   65.136 < 2e-16 ***
## DaysAfterPlanting14    9.722      2.283   4.258 3.89e-05 ***
## DaysAfterPlanting21   11.306      2.283   4.951 2.21e-06 ***
## DaysAfterPlanting28   10.944      2.283   4.793 4.36e-06 ***
## Treatment2       -134.531      3.425 -39.277 < 2e-16 ***
## Treatment3         9.750      3.425   2.847 0.00513 **
## Treatment4         2.719      3.425   0.794 0.42876
## Treatment5        10.719      3.425   3.129 0.00216 **
## Treatment6         8.812      3.425   2.573 0.01119 *
## Treatment7        -2.188      3.425  -0.639 0.52416
## Treatment8         7.750      3.425   2.263 0.02529 *
## Treatment9         2.000      3.425   0.584 0.56028
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.688 on 132 degrees of freedom
## Multiple R-squared:  0.958, Adjusted R-squared:  0.9545
## F-statistic: 273.6 on 11 and 132 DF,  p-value: < 2.2e-16
```

```
anova(lm_model_q3)
```

```
## Analysis of Variance Table
##
## Response: Emergence
##              Df Sum Sq Mean Sq F value    Pr(>F)
## DaysAfterPlanting  3   3116     1039  11.068 1.575e-06 ***
## Treatment          8 279366     34921 372.070 < 2.2e-16 ***
## Residuals        132  12389         94
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 4

Calculate the least square means for Treatment using the emmeans package and perform a Tukey separation with the compact letter display using the cld function. Interpret the results.

From this, we can see that the estimated marginal means of Treatment 2 is lower than all other treatments, although the standard error and df are the same. This contributes to other data demonstrating differences in Treatment 2 when compared to other treatments when considering plant emergence.

```
lsmeans <- emmeans(lm_model_q3, ~Treatment)
result_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details = TRUE)
lsmeans
```

```
##   Treatment emmean   SE  df lower.CL upper.CL
## 1          190.2 2.42 132    185.4    194.9
## 2           55.6 2.42 132     50.8     60.4
## 3          199.9 2.42 132    195.1    204.7
## 4          192.9 2.42 132    188.1    197.7
## 5          200.9 2.42 132    196.1    205.7
## 6          199.0 2.42 132    194.2    203.8
## 7          188.0 2.42 132    183.2    192.8
## 8          197.9 2.42 132    193.1    202.7
## 9          192.2 2.42 132    187.4    196.9
##
## Results are averaged over the levels of: DaysAfterPlanting
## Confidence level used: 0.95
```

```
result_lsmeans
```

```
## $emmeans
##   Treatment emmean   SE  df lower.CL upper.CL .group
## 5          200.9 2.42 132    196.1    205.7  1
## 3          199.9 2.42 132    195.1    204.7  1
## 6          199.0 2.42 132    194.2    203.8  1
```

```

##      8      197.9 2.42 132      193.1      202.7 12
##      4      192.9 2.42 132      188.1      197.7 12
##      9      192.2 2.42 132      187.4      196.9 12
##      1      190.2 2.42 132      185.4      194.9 12
##      7      188.0 2.42 132      183.2      192.8 2
##      2      55.6 2.42 132      50.8      60.4 3
##
## Results are averaged over the levels of: DaysAfterPlanting
## Confidence level used: 0.95
## P value adjustment: tukey method for comparing a family of 9 estimates
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
##
## $comparisons
## contrast      estimate SE df t.ratio p.value
## Treatment7 - Treatment2 132.344 3.43 132 38.638 <0.0001
## Treatment1 - Treatment2 134.531 3.43 132 39.277 <0.0001
## Treatment1 - Treatment7 2.188 3.43 132 0.639 0.9993
## Treatment9 - Treatment2 136.531 3.43 132 39.861 <0.0001
## Treatment9 - Treatment7 4.188 3.43 132 1.223 0.9502
## Treatment9 - Treatment1 2.000 3.43 132 0.584 0.9997
## Treatment4 - Treatment2 137.250 3.43 132 40.071 <0.0001
## Treatment4 - Treatment7 4.906 3.43 132 1.432 0.8832
## Treatment4 - Treatment1 2.719 3.43 132 0.794 0.9969
## Treatment4 - Treatment9 0.719 3.43 132 0.210 1.0000
## Treatment8 - Treatment2 142.281 3.43 132 41.540 <0.0001
## Treatment8 - Treatment7 9.938 3.43 132 2.901 0.0978
## Treatment8 - Treatment1 7.750 3.43 132 2.263 0.3724
## Treatment8 - Treatment9 5.750 3.43 132 1.679 0.7583
## Treatment8 - Treatment4 5.031 3.43 132 1.469 0.8678
## Treatment6 - Treatment2 143.344 3.43 132 41.850 <0.0001
## Treatment6 - Treatment7 11.000 3.43 132 3.212 0.0425
## Treatment6 - Treatment1 8.812 3.43 132 2.573 0.2083
## Treatment6 - Treatment9 6.812 3.43 132 1.989 0.5538
## Treatment6 - Treatment4 6.094 3.43 132 1.779 0.6957
## Treatment6 - Treatment8 1.062 3.43 132 0.310 1.0000
## Treatment3 - Treatment2 144.281 3.43 132 42.124 <0.0001
## Treatment3 - Treatment7 11.938 3.43 132 3.485 0.0187
## Treatment3 - Treatment1 9.750 3.43 132 2.847 0.1120
## Treatment3 - Treatment9 7.750 3.43 132 2.263 0.3724
## Treatment3 - Treatment4 7.031 3.43 132 2.053 0.5099
## Treatment3 - Treatment8 2.000 3.43 132 0.584 0.9997
## Treatment3 - Treatment6 0.938 3.43 132 0.274 1.0000
## Treatment5 - Treatment2 145.250 3.43 132 42.406 <0.0001
## Treatment5 - Treatment7 12.906 3.43 132 3.768 0.0074
## Treatment5 - Treatment1 10.719 3.43 132 3.129 0.0535
## Treatment5 - Treatment9 8.719 3.43 132 2.545 0.2204
## Treatment5 - Treatment4 8.000 3.43 132 2.336 0.3288
## Treatment5 - Treatment8 2.969 3.43 132 0.867 0.9943
## Treatment5 - Treatment6 1.906 3.43 132 0.557 0.9998
## Treatment5 - Treatment3 0.969 3.43 132 0.283 1.0000
##

```

```
## Results are averaged over the levels of: DaysAfterPlanting
## P value adjustment: tukey method for comparing a family of 9 estimates
```

Question 4

The provided function lets you dynamically add a linear model plus one factor from that model and plots a bar chart with letters denoting treatment differences. Use this model to generate the plot shown below. Explain the significance of the letters.